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EARPHONES

Abstract

The present disclosure relates to an earphone, including: an ear hook and a core module connected to the ear hook, the core module including an outer shell, a decorative cover, and a touch module, the outer shell including an outer surface away from an ear of a user when the earphone is in a wearing state, the decorative cover being disposed on the outer surface of the outer shell and configured to form a touch positioning region for the user to perform touch positioning; a positioning protrusion being disposed on the outer shell and/or the decorative cover, the positioning protrusion being located in the touch positioning region and protruding from an outer surface of the decorative cover away from the outer shell; the touch module being disposed on the outer shell and including a touch detection region that at least partially overlaps with the touch positioning region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2023/131100, filed on Nov. 10, 2023, which claims priority to Chinese application No. 202310956970.8, filed on Jul. 28, 2023, the entire contents of each of which are incorporated herein by reference.

TECHNICAL FIELD
[0002] This application relates to the technical field of electronic devices, and in particular, to an earphone.
BACKGROUND
[0003] With the widespread adoption of electronic devices, electronic devices have become indispensable tools for social interaction and entertainment in daily life. As a result, users are placing increasingly higher demands on the electronic devices. Electronic devices such as earphones and smart glasses have also been widely used in people's daily lives. The earphones and smart glasses are often used in conjunction with terminals such as smartphones and computers to provide users with an enhanced auditory experience. Current earphones are typically provided with a touch control region on the outer shell, allowing users to control the earphones by touching this region. However, since the earphones are worn on the user's ears during use, it is often difficult for the user to accurately locate and touch the touch control region during operation.
SUMMARY
[0004] The present application provides an earphone, comprising an ear hook and a core module.
[0005] The core module is connected to the ear hook, the core module including an outer shell, a decorative cover, and a touch module, the outer shell including an outer surface away from an ear of a user when the earphone is in a wearing state, the decorative cover being disposed on the outer surface of the outer shell and configured to form a touch positioning region for the user to perform touch positioning; a positioning protrusion being disposed on the outer shell and/or the decorative cover, the positioning protrusion being located in the touch positioning region and protruding from an outer surface of the decorative cover away from the outer shell; the touch module being disposed on the outer shell and including a touch detection region that at least partially overlaps with the touch positioning region.
[0006] Through the above configuration, based on the touch positioning region providing a first touch navigation positioning, setting the positioning protrusion to protrude from the outer surface of the decorative cover that is away from the outer shell facilitates the user to touch the positioning protrusion when touching the outer shell, so that the positioning protrusion can provide a second touch navigation positioning based on the decorative cover to facilitate the user to more accurately locate the touch detection region by touching the positioning protrusion, improving the convenience of touch operation. Moreover, the decorative cover can bring a decorative unique

visual effect to the earphone, and compounding (at least partially overlapping) with the touch detection region based on the unique visual effect can eliminate the need to additionally set up a detection region for touch, making the earphone smaller in size. Due to the visual effect of the decorative cover, it is also easy for the user to be aware of the location of the touch detection region, reducing the user's learning cost and improving the user's convenience of touch operation.

[0007] In some embodiments, at least a portion of an edge of the decorative cover is exposed on the outer surface of the outer shell to enclose the touch positioning region.

[0008] In some embodiments, the edge of the decorative cover is configured as a ring and is completely exposed on the outer surface of the outer shell to enclose the touch positioning region; and the edge of the decorative cover surrounds the positioning protrusion.

[0009] In some embodiments, a positioning groove is disposed on the outer surface of the outer shell, and the decorative cover is accommodated in the positioning groove through a groove opening of the positioning groove.

[0010] In some embodiments, the positioning groove and the decorative cover match in shape and size; and/or, a seam between an edge of the decorative cover and an edge of the groove opening of the positioning groove is exposed on the outer surface, and the seam surrounds the positioning protrusion.

[0011] In some embodiments, the positioning protrusion protrudes from the outer surface of the outer shell, the decorative cover is provided with a positioning hole, and the positioning protrusion passes through the positioning hole; or, the positioning protrusion protrudes from the outer surface of the decorative cover and is integrally formed with the decorative cover.

[0012] In some embodiments, the outer shell has a length direction, a width direction, and a thickness direction that are perpendicular to each other, and the outer shell is stacked on the ear of the user along the thickness direction when the earphone is in the wearing state; a ratio of a dimension of the decorative cover to a dimension of the outer shell along the length direction is greater than or equal to 0.5 and less than or equal to 0.8; and a ratio of a dimension of the decorative cover to a dimension of the outer shell along the width direction is greater than or equal to 0.6 and less than or equal to 0.95.

[0013] In some embodiments, the outer shell includes a connection end connected to the ear hook and a free end away from the connection end along the length direction, and the decorative cover is closer to the connection end.

[0014] In some embodiments, the outer shell is bent along the thickness direction toward the outer surface of the outer shell so that the free end is closer to an ear hole of the ear of the user than the connection end.

[0015] In some embodiments, the outer shell has a length direction, a width direction, and a thickness direction that are perpendicular to each other, and the outer shell is stacked on the ear of the user along the thickness direction when the earphone is in the wearing state; and the positioning protrusion is configured as a strip, and an extension direction of the positioning protrusion intersects with the length direction and the width direction, respectively.

[0016] In some embodiments, a roughness of the outer surface of the decorative cover is greater than a roughness of the outer surface of the outer shell; and/or a color of the outer surface of the decorative cover is different from a color of the outer surface of the outer shell; and/or a material of the decorative cover is different from a material of the outer shell.

[0017] In some embodiments, the touch module includes a touch metal pattern and a circuit board,

[0018] the circuit board is disposed inside the outer shell, the touch metal pattern is disposed inside the outer shell or sandwiched between the decorative cover and the outer shell, the touch metal pattern is electrically connected to the circuit board, and the touch metal pattern is configured to form the touch detection region.

[0019] In some embodiments, the touch metal pattern is further configured as a radio frequency antenna of the earphone.

[0020] In some embodiments, the outer shell has a length direction, a width direction, and a thickness direction that are perpendicular to each other, and the outer shell is stacked on the ear of the user along the thickness direction when the earphone is in the wearing state; the outer shell includes a connection end connected to the ear hook and a free end away from the connection end along the length direction, and the outer shell is bent along the thickness direction toward the outer surface of the outer shell, so that the free end is closer to an ear hole of the ear of the user than the connection end, and the decorative cover is closer to the connection end.
[0021] In some embodiments, the outer surface of the outer shell and the outer surface of the decorative cover together form an outer surface of the core module; the outer surface of the core module is configured as an outwardly arched shape; at least two sound pickup holes are disposed on the outer surface of the core module; the two sound pickup holes are located on two opposite slopes of the outer surface of the core module, respectively; and/or, the touch detection region is closer to the connection end than the positioning protrusion, or the touch detection region surrounds the positioning protrusion.
[0022] In some embodiments, the positioning protrusion is configured as a strip, and the two sound pickup holes are located on two sides of the positioning protrusion, respectively.
[0023] In some embodiments, the free end has a first reference position farthest from the connection end along the length direction, and the decorative cover has a second reference position closest to the first reference position along the length direction; the outer shell includes an arcuate transition region between the first reference position and the second reference position; and from the second reference position to the first reference position, the arcuate transition region gradually approaches the ear of the user along the thickness direction when the earphone is in the wearing state.
[0024] In some embodiments, the outer shell forms an accommodation space, and the core module is provided with two sound pickup holes, the accommodation space is in flow communication with the external environment through the two sound pickup holes, one of the two sound pickup holes passes through the outer shell and is located at a periphery of the decorative cover, and the other one of the two sound pickup holes passes through the outer shell and the decorative cover; and the core module includes two microphones, and the two microphones are spaced apart in the accommodation space and correspond one-to-one with the two sound pickup holes.
[0025] In some embodiments, a distance between the two sound pickup holes is greater than or equal to a dimension of the decorative cover along an arrangement direction of the two microphones; and/or, the outer shell has a length direction and a width direction that are perpendicular to each other, the outer shell has a connection end and a free end along the length direction; the connection end is connected to the ear hook; the arrangement direction of the two microphones intersects with the length direction and the width direction, respectively; and the sound pickup hole that passes through the decorative cover is closer to the connection end and an edge of the decorative cover away from the ear hook along the width direction than the sound pickup hole located at the periphery of the decorative cover; and/or, the touch detection region is located between the two sound pickup holes.
[0026] In some embodiments, the outer shell forms an accommodation space, and the core module is provided with a sound pickup hole passing through the outer shell, the accommodation space is in flow communication with the external environment though the sound pickup hole; the core module includes a microphone and a sealing component, and the microphone is disposed in the accommodation space and corresponds to the sound pickup hole to pick up sound of the external environment through the sound pickup hole; and at least a portion of the sealing component is disposed between the outer shell and the microphone and is configured to isolate a sound pickup path from the accommodation space.
[0027] In some embodiments, the core module further includes a fixing plate, the fixing plate is accommodated in the accommodation space and includes a through hole, and the microphone is

fixed to one side of the fixing plate and covers the through hole; and at least a portion of the sealing component is disposed in the accommodation space and abuts between the outer shell and a side of the fixing plate away from the microphone, and enables the through hole and the sound pickup hole to be in flow communication with each other.
[0028] In some embodiments, the microphone passes through the outer shell and is located at a periphery of the decorative cover; the sealing component includes a sealing gasket and a mesh component, and the sealing gasket and the mesh component are stacked and disposed in the accommodation space between the fixing plate and the outer shell.
[0029] In some embodiments, the sealing gasket is stacked on the fixing plate and is closer to the through hole than the mesh component; and the mesh component is located between the sealing gasket and the outer shell and is closer to the sound pickup hole than the sealing gasket.
[0030] In some embodiments, the microphone passes through the outer shell and the decorative cover; and the sealing component includes a sealing gasket and at least one mesh component, the sealing gasket is disposed in the accommodation space between the outer shell and the fixing plate and directly abuts the outer shell and the fixing plate, and the least one mesh component is disposed between the outer shell and the decorative cover.
[0031] In some embodiments, the least one mesh component includes two mesh components, one of the two mesh components is disposed between the outer shell and the decorative cover, and the other one of the two mesh components is stacked with the sealing gasket and located between the outer shell and the fixing plate.
[0032] In some embodiments, the least one mesh component includes an annular double-sided adhesive and a mesh, the annular double-sided adhesive is stacked with the mesh and bonded to the mesh, and the mesh faces the through hole and the sound pickup hole.
[0033] In some embodiments, the least one mesh component includes two layers of annular double-sided adhesive and one layer of mesh, and the mesh is stacked and bonded between the two layers of annular double-sided adhesive; or, the least one mesh component includes two layers of annular double-sided adhesive and two layers of mesh, the two layers of annular double-sided adhesive are spaced apart, one layer of the mesh is stacked and bonded between the two layers of annular double-sided adhesive, and one layer of the annular double-sided adhesive is stacked between the two layers of mesh.
[0034] In some embodiments, the fixing plate is a circuit board.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0035] To more clearly illustrate the technical solutions in the embodiments of the present disclosure, the following will briefly introduce the accompanying drawings that need to be used in the description of the embodiments, and it will be obvious that the accompanying drawings in the following description are only some of the embodiments of the present disclosure, and other attachments can be obtained based on these drawings without creative labor to a person of ordinary skill in the art.
[0036] FIG. **1** is a schematic diagram illustrating an anterior contour of a user's ear;
[0037] FIG. **2** is a schematic diagram illustrating an overall structure of an earphone according to embodiments of the present disclosure;
[0038] FIG. **3** is a schematic diagram illustrating an exemplary internal structure of an earphone according to embodiments of the present disclosure;
[0039] FIG. **4** is a schematic diagram illustrating an exemplary structure of a first shell of an earphone according to some embodiments of the present disclosure;
[0040] FIG. **5** is a schematic diagram illustrating a cross-sectional view along A-A section plane in

FIG. **2**;

[0041] FIG. **6** is a schematic diagram illustrating positions of upper reference points on a first shell and a second shell in FIG. **5**;

[0042] FIG. **7** is a schematic diagram illustrating positions of other reference points on a first shell and a second shell in FIG. **5**;

[0043] FIG. **8** is a schematic diagram illustrating an exemplary structure of an earphone from a perspective according to embodiments of the present disclosure;

[0044] FIG. **9** is a schematic diagram illustrating an exemplary structure of a charging box used in conjunction with an earphone according to embodiments of the present disclosure;

[0045] FIG. **10** is a schematic diagram illustrating an exemplary structure of an earphone from another perspective according to embodiments of the present disclosure;

[0046] FIG. **11** is a schematic diagram illustrating an exemplary structure of an earphone according to embodiments of the present disclosure;

[0047] FIG. **12** is a schematic diagram illustrating another exemplary structure of an earphone according to embodiments of the present disclosure;

[0048] FIG. **13** is a schematic diagram illustrating another exemplary structure of an earphone according to embodiments of the present disclosure;

[0049] FIG. **14** is a schematic diagram illustrating another exemplary structure of an earphone according to embodiments of the present disclosure;

[0050] FIG. **15** is a schematic diagram illustrating an exemplary structure of assemblies of an earphone according to embodiments of the present disclosure;

[0051] FIG. **16** is a schematic diagram illustrating an exemplary structure of other assemblies of an earphone according to embodiments of the present disclosure;

[0052] FIG. **17** is a schematic diagram illustrating a front view of the earphone shown in FIG. **2**;

[0053] FIG. **18** is a schematic diagram illustrating a cross-sectional view of the earphone shown in FIG. **17** along M-M section line;

[0054] FIG. **19** is a schematic diagram illustrating an enlarged view of a region C of the earphone shown in FIG. **17**;

[0055] FIG. **20** is a schematic diagram illustrating an enlarged view of region B of the earphone shown in FIG. **17**;

[0056] FIG. **21** is a schematic diagram illustrating a three-dimensional structure of an earphone according to embodiments of the present disclosure;

[0057] FIG. **22** is a schematic diagram illustrating a cross-sectional view of the earphone shown in FIG. **21**;

[0058] FIG. **23** is a schematic diagram illustrating an exemplary exploded structure of a core module of the earphone shown in FIG. **21**;

[0059] FIG. **24** is a schematic diagram illustrating an exemplary structure of a basket shown in FIG. **23** assembled in a first accommodation cavity;

[0060] FIG. **25** is a schematic diagram illustrating an enlarged structure of region A shown in FIG. **24**;

[0061] FIG. **26** is a schematic diagram illustrating another enlarged structure of the region A shown in FIG. **24**;

[0062] FIG. **27** is a schematic diagram illustrating a three-dimensional structure of a loudspeaker shown in FIG. **23**;

[0063] FIG. **28** is a schematic diagram illustrating an assembled structure of a loudspeaker and a bracket shown in FIG. **23**; and

[0064] FIG. **29** is a schematic diagram illustrating a three-dimensional structure of the bracket shown in FIG. **23**.

DETAILED DESCRIPTION

[0065] The present disclosure is described in further detail below in conjunction with the accompanying drawings and embodiments. In particular, it is noted that the following embodiments are only used to illustrate the present disclosure, but do not limit the scope of the present disclosure. Similarly, the following embodiments are only part of the embodiments of the present disclosure rather than all of the embodiments, and all other embodiments obtained by a person of ordinary skill in the art without creative labor fall within the scope of protection of the present disclosure.

[0066] References to “embodiments” in the present disclosure mean that particular features, structures, or characteristics described in conjunction with embodiments may be included in at least one embodiment of the present disclosure. It is understood by a person of ordinary skill in the art, both explicitly and implicitly, that the embodiments described in the present disclosure may be combined with other embodiments.

[0067] In conjunction with FIG. **1**, a user's ear **500** includes physiological parts such as an external ear canal **501**, a concha cavity **502**, a cymba conchae **503**, a triangular fossa **504**, an antihelix **505**, a scapha **506**, a helix **507**, and a tragus **508**. Although the external ear canal **501** has a certain depth and extends to a tympanic membrane of the ear, for ease of description and in conjunction with FIG. **1**, when not otherwise specified in the present disclosure, the external ear canal **501** specifically refers to an entrance (i.e., the ear hole) of the external ear canal **501** away from the tympanic membrane. Furthermore, the physiological parts such as the concha cavity **502**, the cymba conchae **503**, the triangular fossa **504** have certain volumes and certain depths. The concha cavity **502** is in direct connection with the external ear canal **501**, which means that the ear hole can be simply regarded as being located at the bottom of the concha cavity **502**.

[0068] Furthermore, different users may have individual differences that result in different shapes, sizes, and other dimensional differences in the ear. In order to facilitate the description and to minimize (or even eliminate) the individual differences between different users, a simulator with a head and (left and right) ears can be made based on the ANSI: S3.36, S3.25 and IEC: 60318-7 standards, e.g., GRAS 45BC KEMAR. Accordingly, in the present disclosure, the terms “a user wears an earphone,” “when an earphone is in a wearing state,” “in a wearing state,” etc., refer to that an earphone described in the present disclosure is worn on the ear of the simulator. It is understandable that it is precisely because of the individual differences between different users that the earphone may differ somewhat from when it is worn by different users and when it is worn on the simulator of the ear, but such differences should be tolerated.

[0069] Referring to FIG. **2** to FIG. **5**, an earphone **100** may include a core module **1** and an ear hook **2**, and the core module **1** may include a shell assembly **10** and a loudspeaker **13**. The ear hook **2** may be configured as a hook, and the earphone **100** may be worn through the ear hook **2** at a position near but not blocking the external ear canal of a user's ear. The ear hook **2** may be connected to the shell assembly **10**. In a wearing state, the shell assembly **10** is located at the front side of the ear, and at least a portion of the ear hook **2** is located at the rear side of the user's ear to allow the earphone **100** to be hooked on the ear. Further, the shell assembly **10** may be used to form an accommodation space **101**. The loudspeaker **13** may be disposed in the accommodation space **101**, and the shell assembly **10** is provided with a sound outlet hole for transmitting air-conducted sound provided by the loudspeaker **13** to the external ear canal of the user's ear.

[0070] For ease of describing the shape of the shell assembly **10**, the present disclosure defines that the shell assembly **10** extends along a first direction X and a second direction Y, and the first direction X may be orthogonal to the second direction Y (see FIG. **5**). The first direction X refers to a length direction of the shell assembly **10**, and the second direction Y refers to a thickness direction of the shell assembly **10**.

[0071] In the wearing state, the shell assembly **10** is stacked on the front side of the user's ear along the second direction Y. The shell assembly **10** further has a first end **102** and a second end **103** disposed opposite each other along the first direction X. In some embodiments, the first end **102** refers to a portion facing the user's concha cavity **502** during wearing, and the second end **103** refers to a portion away from the user's concha cavity **502** compared to the first end **102**. In some embodiments, the first end **102** refers to a portion that is closer to the rear side of the user's ear during wearing, and the second end **103** refers to a portion away from the rear side of the user's ear compared to the first end **102**. A portion of the second end **103** may be connected to the ear hook **2**, which may be referred to as the connection end **103** hereinafter, and the first end **102** may also be referred to as a free end **102** hereinafter.

[0072] In some embodiments, at least a portion of the first end **102** of the shell assembly **10** extends into the concha cavity **502** of the user's ear and makes the external ear canal of the user's ear in an open state. The user's external ear canal being in an open state refers to that the shell assembly **10** does not completely close off the external ear canal, or the shell assembly **10** does not block the external ear canal so that the external ear canal may be in flow communication with the external environment. Alternatively, due to the individual differences between different users, when the earphone **100** is worn by different users, the shell assembly **10** may block the external ear canal to some extent, but the external ear canal is still not completely blocked, and the external ear canal can still be in flow communication with the external environment.

[0073] In some embodiments, referring to FIG. **5**, FIG. **6**, and FIG. **7**, in the second direction Y, an outer wall surface LS**2** of the shell assembly **10** of the core module **1** facing the user's ear when the earphone is in the wearing state is configured in an arch shape away from the user's ear. In such cases, the design of the shell assembly **10** can fit the shape of the user's ear more closely, which reduces the pressure of the shell assembly **10** on the user's ear, especially on the tragus **508**. In addition, the restriction on the extension direction of the plane can be removed, which is conducive to the first end **102** of the shell assembly **10** to more smoothly extend into the user's concha cavity **502**, thereby improving the comfort and stability of wearing.

[0074] The structural design of the outer wall surface LS**2** of the shell assembly **10** facing the user's ear in a wearing state can have an impact on the structure of the entire shell assembly **10**, which in turn can affect the comfort of the user's wearing. For example, the structure of the outer wall surface LS**2** affects the position of the first end **102** of the shell assembly **10**, and if the outer wall surface LS**2** results in the extension of the shell assembly **10** being too short, it may cause the first end **102** to be difficult to extend into the user's concha cavity **502**, or it may cause that when the first end **102** of the shell assembly **10** extends into the user's concha cavity **502**, the outer wall surface LS**2** cannot avoid the user's tragus **508** well, leading to excessive pressure on the user's tragus **508**. As another example, if the outer wall surface LS**2** results in the extension of the shell assembly **10** being too long, the pressure of the first end **102** on the antihelix **505** and the concha cavity **502** may be increased, or it may cause that the outer wall surface LS**2** cannot avoid the user's tragus **508** well, leading to excessive pressure on the user's tragus **508**.

[0075] According to FIG. **6**, optionally, in the first direction X, the first end **102** has a first reference point C**1** that is farthest away from the second end **103**. In some embodiments, the first reference point C**1** may be a point of the first end **102** that is farthest from the second end **103** along the first direction X. In some embodiments, the first reference point C**1** may be a point on the first end **102** that is farthest from the second end **103** along the tangential direction of the arch of the outer wall surface LS**2**. To prevent the arch-shaped outer wall surface LS**2** from causing a portion of the shell assembly **10** that is closer to the first end **102** to be too short or too long, which results in that the first end **102** cannot fit the concha cavity **502** well and reduces the user's wearing comfort, a distance L**10** between an apex G**1** of the outer wall surface LS**2** and the first reference point C**1** along the first direction X is in a range from 14 mm to 20 mm. The apex G**1** of the outer wall surface LS**2** refers to a point with the greatest recessed depth in an arched region of the arch shape, or a point farthest from the user's ear in the arched region. The distance between the apex G**1** and the first reference point C**1** along the first direction X refers to the length of the projection of the line connecting the apex G**1** and the first reference point C**1** along the first direction X.

[0076] Further, to make the earphone **100** adapt to the shape of the user's ear, the distance L**10** between the apex G**1** of the outer wall surface LS**2** and the first reference point C**1** along the first direction X may be more targeted. In some embodiments, in the case where the user's ear is small, to prevent the arch-shaped outer wall surface LS**2** from causing the portion of the shell assembly **10** that is closer to the first end **102** to be too long, which compresses the concha cavity **502** of the user, the distance L**10** between the apex G**1** of the outer wall surface LS**2** and the first reference point C**1** along the first direction X may be in a range from 14 mm to 16 mm. In some embodiments, in the case where the user's ear is large, to prevent the arch-shaped the outer wall surface LS**2** from causing the portion of the shell assembly **10** that is closer to the first end **102** being too short, which results in that the first end **102** cannot abut against the concha cavity **502** well and leads to the problem of unstable wearing, the distance L**10** between the apex G**1** of the outer wall surface LS**2** and the first reference point C**1** along the first direction X may be in a range from 18 mm to 20 mm. Specifically, the distance L**10** between the apex G**1** of the outer wall surface LS**2** and the first reference point C**1** along the first direction X may be 14.1 mm, 15.2 mm, 16.5 mm, 17.7 mm, 18.6 mm, or 19.9 mm.

[0077] Optionally, in the second direction Y, the outer wall surface LS**2** of the shell assembly **10** facing the user's ear has a second reference point C**2** adjacent to the second end **103** and closest to the user's ear when the earphone is in the wearing state. In some embodiments, the second reference point C**2** may be a portion of the outer wall surface LS**2** that is close to the second end **103** and a point that is farthest from the apex G**1** or a decorative cover (described later) along the second direction Y. To prevent the second end from being unable to stably support on the user's face when the first end **102** extends into the concha cavity **502**, resulting in reduced wearing stability, a distance L**12** between the second reference point C**2** and the first reference point C**1** along the first direction X is in a range from 20 mm to 31 mm. Alternatively, the second reference point C**2** may also be the lowest point on the arch-shaped outer wall surface LS**2** of the shell assembly **10** facing the user's ear. Optionally, the second reference point C**2** is configured to abut the face skin located on the front side of the tragus **508** of the user's ear in the wearing state.

[0078] Further, to make the earphone **100** adapt to the shape of the user's ear, the distance L**1** between the second reference point C**2** and the first reference point C**1** along the first direction X may be more targeted. In some embodiments, in the case where the user's ear is small, to prevent the first end **102** from being unable to abut against the concha cavity **502** well when the second reference point C**2** is supported on the tragus, the distance L**1** between the second reference point C**2** and the first reference point C**1** along the first direction X may be in a range from 20 mm to 24 mm. In some embodiments, in the case where the user's ear is large, to prevent the region where the second reference point C**2** is located from being supported on the tragus and over-pressing the user's tragus, the distance L**12** between the second reference point C**2** and the first reference point C**1** along the first direction X may be in a range from 26 mm to 31 mm. Specifically, the distance L**12** between the second reference point C**2** and the first reference point C**1** along the first direction X may be 21.9 mm, 23.4 mm, 25 mm, 26.4 mm, or 28.02 mm.

[0079] In this case, the earphone **100** can be more stably supported on the face skin located on the front side of the user's tragus **508** through the second reference point C**2**, and when the earphone **100** is supported on the face skin located on the front side of the user's tragus **508** through the second reference point C**2**, the first end **102** can extend into the concha cavity **502** without excessively compressing the antihelix **505**.

[0080] Optionally, a distance H**20** between the apex G**1** of the outer wall surface LS**2** and the second reference point C**2** along the second direction Y is in a range from 1.3 mm to 2.2 mm. Further, to make the earphone **100** adapt to the shape of the user's ear, the distance H**20** between the apex G**1** of the outer wall surface LS**2** and the second reference point C**2** along the second direction Y may be more targeted. Specifically, the distance H**20** between the apex G**1** of the outer wall surface LS**2** and the second reference point C**2** along the second direction Y may be 1.4 mm, 1.5 mm, 1.6 mm, 1.7 mm, 1.8 mm, 1.9 mm, 2.0 mm, or 2.1 mm. In such a case, when the earphone **100**

is supported on the face skin located on the front side of the user's tragus **508** through the second reference point C**2**, the outer wall surface LS**2** does not excessively compress the user's tragus **508**.
[0081] Typically, limited by the volume of the concha cavity **502**, if the first end **102** is too large or too thick, it may be difficult for the first end **102** to extend into the concha cavity **502**. If the first end **102** is too small, it is difficult to utilize the internal space.
[0082] Optionally, in the second direction Y, the outer wall surface LS**2** of the shell assembly **10** facing the user's ear has a third reference point C**3** that is adjacent to the first end **102** and closest to the user's ear when the earphone is in the wearing state. In some embodiments, the third reference point C**3** may be a portion of the outer wall surface LS**2** that is close to the first end **102** and a point that is farthest from the apex G**1** along the second direction Y. In some embodiments, the third reference point C**3** may also be another lowest point on the arch-shaped outer wall surface LS**2** of the shell assembly **10** facing the user's ear.
[0083] Optionally, a distance L**13** between the third reference point C**3** and the first reference point C**1** along the first direction X is in a range from 3 mm to 5 mm. Further, to make the earphone **100** fit the shape of the user's ear and increase the internal space of the shell assembly **10**, the distance L**13** between the third reference point C**3** and the first reference point C**1** along the first direction X should be as small as possible without overly compressing the concha cavity **502**. For example, the distance L**13** between the third reference point C**3** and the first reference point C**1** along the first direction X may be in a range from 3 mm to 3.5 mm, in a range from 3.51 mm to 4 mm, in a range from 4.01 mm to 4.5 mm, or in a range from 4.51 mm to 5 mm. For example, the distance L**13** between the third reference point C**3** and the first reference point C**1** along the first direction X may be 3.45 mm, 3.8 mm, 4.29 mm, or 4.53 mm. In this case, the first end **102** of the shell assembly **10** can extend into the concha cavity **502** relatively easily, and it is relatively easy to utilize the internal space (e.g., to facilitate the assembly of the loudspeaker **13**).
[0084] Considering that if the apex G**1** of the outer wall surface LS**2** is recessed too little toward the inner side of the shell assembly **10**, the outer wall surface LS**2** may excessively compress the tragus **508** in the wearing state. If the apex G**1** of the outer wall surface LS**2** is recessed too much toward the inner side of the shell assembly **10**, the curvature of the outer wall surface LS**2** and the corresponding inner wall surface LS**1** (described later) of the shell assembly **10** may be too large, which is not conducive to the layout of various components inside the shell assembly **10**.
[0085] Optionally, a distance H**30** between the apex G**1** of the outer wall surface LS**2** and the third reference point C**3** along the second direction Y is in a range from 2 mm to 3 mm. For example, the distance H**30** between the apex G**1** of the outer wall surface LS**2** and the third reference point C**3** along the second direction Y may be 2.1 mm, 2.2 mm, 2.3 mm, 2.4 mm, 2.5 mm, 2.6 mm, 2.7 mm, 2.8 mm, or 2.9 mm. In this case, the outer wall surface LS**2** will not excessively compress the tragus **508** during wearing, and the outer wall surface LS**2** of the shell assembly **10** may not bend excessively.
[0086] Currently, the battery **14** and the loudspeaker **13** in the earphone **100** are usually located in two separate shells, which requires a long wire to connect the battery **14** and the loudspeaker **13**. However, during the assembly of the earphone **100** and the use of the user, the wire is often broken due to external forces, resulting in the earphone **100** not being able to be used properly, and separating the battery **14** and the loudspeaker **13** into two separate shells also increases the size of the earphone **100** and reduces the wearing experience of the user.
[0087] In some embodiments, the earphone **100** further includes a battery **14**, which may be configured to supply power to the loudspeaker **13**. In the wearing state, since the first end **102** of the shell assembly **10** is closer to the external ear canal of the user's ear than the second end **103**, the loudspeaker **13** may be located closer to the first end **102** within the shell assembly **10**, and the battery **14** may be located closer to the second end **103** within the shell assembly **10**. In this case, it is more favorable for the air-conducted sound generated by the loudspeaker **13** to be delivered to the external ear canal of the user, ensuring the user's listening effect.

[0088] In some embodiments, the loudspeaker **13** and the battery **14** may be spaced apart in the accommodation space **101** formed by the shell assembly **10** along the first direction X. In such a case, the loudspeaker **13** and the battery **14** may be assembled in the shell assembly **10** at the same time, thereby improving the assembly efficiency, reducing the length of the wire, and allowing the wire to be set within the shell assembly **10** and not easily damaged.

[0089] Further, a partition wall **104** may be disposed in the accommodation space **101** of the shell assembly **10**, as shown in FIG. **6**. The partition wall **104** may be used to separate the loudspeaker **13** and the battery **14** apart to avoid damaging the loudspeaker **13** in the event of leakage of the battery **14**, etc. In some embodiments, the partition wall **104** may be integrally molded with the shell assembly **10**, which may improve the consistency of the shell assembly **10**. In some embodiments, the partition wall **104** may be separately molded and mounted within the shell assembly **10** by snap-fit, glue, or the like, to reduce the difficulty of processing the shell assembly **10**. Referring to FIG. **4** and FIG. **5**, for example, an inner wall surface LS**1** of the shell assembly **10** facing the user's ear is provided with the partition wall **104** that is adjacent to the apex G**1** of the outer wall surface LS**2**, and the loudspeaker **13** is disposed within an accommodation space formed by the partition wall **104** and the first end **102**, and the battery **14** is disposed within an accommodation space formed by the partition wall **104** and the second end **103**.

[0090] In some embodiments, the loudspeaker **13** has a first axis ZX**1** and the battery **14** has a second axis ZX**2**, with each of the first axis ZX**1** and the second axis ZX**2** having a positive direction pointing toward the user's ear in the wearing state. On a reference plane defined by the first direction X and the second direction Y, a projection of the positive direction of the first axis line ZX**1** and a projection of the second axis line ZX**2** on the reference plane intersect, i.e., the loudspeaker **13** and the battery **14** are set at an inclination relative to each other. Such a setting can improve the space utilization rate of the shell assembly **10** along the first direction X, ensure that the shell assembly **10** can accommodate the various electronic components without increasing its size, and prevent the shell assembly **10** from being too large to affect the wearing experience of the user. The first axis ZX**1** may be parallel to the vibration direction of a vibration diaphragm of the loudspeaker **13** and passes through the center of gravity of the loudspeaker **13**, or the geometric centers or centroids of end surfaces (e.g., circular end surface, runway-shaped end surface) of the loudspeaker **13**. The second axis ZX**2** refers to the axis of symmetry that passes through the battery **14**, or an axis passing through the geometric centers or centroids of two end surfaces of the battery **14**.

[0091] Further, referring to FIG. **5**, an intersection angle **0** between a projection of the positive direction of the first axis ZX**1** on the reference plane and a projection of the positive direction of the second axis ZX**2** on the reference plane is an acute angle. To prevent an inclination angle between the battery **14** and the loudspeaker **13** from being too large while reducing the size of the shell assembly **10** along the first direction X, so as to avoid the shell assembly **10** from becoming larger along the second direction Y, the intersection angle θ between the projection of the positive direction of the first axis ZX**1** on the reference plane and the positive direction of the second axis ZX**2** on the reference plane may be in a range from 10° to 14° or a range from 11° to 14°. For example, the intersection angle θ may be 10°, 10.5°, 11°, 11.5°, 12°, 12.5°, 13°, 13.5°, or 14°. In some embodiments, the first axis ZX**1** and the second axis ZX**2** may be parallel to the reference plane, in which case the angle between the positive direction of the first axis ZX**1** and the positive direction of the second axis ZX**2** may be equal to the intersection angle θ.

[0092] In some embodiments, a radial dimension of the loudspeaker **13** is larger than a radial dimension of the battery **14** on a reference plane defined by the first direction X and the second direction Y, the second direction Y has a positive direction pointing toward the user's ear, and an intersection angle θ**1** between a projection of the positive direction of the first axis ZX**1** on the reference plane and the positive direction of the second direction Y is larger than an intersection angle θ**2** between a projection of the positive direction of the second axis ZX**2** and the positive direction of the second direction Y. The second axis ZX**2** may be parallel to the second direction Y, thereby lowering the assembly difficulty of the battery **14**. The radial dimension of the loudspeaker **13** refers to the length of the cross-section of the loudspeaker **13** on the reference plane along the direction perpendicular to the first axis ZX**1**. The radial dimension of the battery **14** refers to the length of the cross-section of the battery **14** on the reference plane along a direction perpendicular to the second axis ZX**2**.

[0093] In such cases, since the radial dimension of the loudspeaker **13** is larger than the radial dimension of the battery **14**, if the loudspeaker **13** and the battery **14** are arranged side-by-side along the first direction X, the space occupied by the loudspeaker **13** along the first direction X is relatively large. Therefore, when the loudspeaker **13** is arranged inclined with respect to the second direction Y as compared to the battery **14**, the influence on the change degree of the projection length of the shell assembly **10** along the first direction X is greater, which facilitates the reduction of the length of the shell assembly **10** along the first direction X.

[0094] The earphone **100** further includes a circuit board **15**, the circuit board **15** being disposed in the accommodation space **101** formed by the shell assembly **10** and located on a side of the loudspeaker **13** and the battery **14** away from the user's ear. In some embodiments, the earphone **100** includes one circuit board **15**, to which wires of the battery **14** and the loudspeaker **13** are connected for control, and a projection of the circuit board **15** along the second direction Y at least partially overlaps with the loudspeaker **13** and the battery **14**. As shown in FIG. **5**, the circuit board **15** may be arranged along the first direction X, which increases space utilization and reduces production costs while ensuring the performance of the loudspeaker **13** and the battery **14**. In some embodiments, the earphone **100** includes two circuit boards **15**, the two circuit boards **15** are electrically connected to the battery **14** and the loudspeaker **13**, respectively, and the axes of the two circuit boards **15** are parallel to the first axis ZX**1** of the loudspeaker **13** and the second axis ZX**2** of the battery **14**, respectively, which further reduces the size of the shell assembly **10**.

[0095] In some embodiments, the shell assembly **10** includes a first shell **111** and a second shell **112** that match each other along the second direction Y. The loudspeaker **13** and the battery **14** may be fixed to the first shell **111**, in the wearing state, the first shell **111** is closer to the user's ear than the second shell **112**, and the circuit board **15** is fixed to the second shell **112**, which is convenient to assemble the earphone **100** facilitates the disassembly and repair of the earphone **100**. The circuit board **15** may be fixed to the second shell **112** of the shell assembly **10** by hot melting. Specifically, a plurality of hot melting posts may be provided on the second shell **112**, and a plurality of through holes may be provided on the circuit board **15**. During assembly, the through holes on the circuit board **15** may cooperate with the hot melting posts on the second shell **112**, and the circuit board **15** may be fixed to the second shell through hot melting.

[0096] See FIG. **5** and FIG. **8**, the earphone **100** further includes a charging interface **16** disposed on the first shell **111**, and a magnetic suction element **17** disposed between the battery **14** and the first shell **111** along the second axis direction ZX**2**. Disposing the magnetic suction element **17** between the battery **14** and the first shell **111** can make full use of the accommodation space **101** and increase the space utilization rate. In this case, the earphone **100** may be used with a charging box **900** having a magnetic charging function.

[0097] See FIG. **9**, the charging box **900** may include a lower shell assembly **910** and an upper shell assembly **920**. The lower shell assembly **910** may be provided with two profiling recesses **911** for accommodating the earphones **100**, respectively, and the upper shell assembly **920** may cover the lower shell assembly **910** to close the profiling recesses **911**.

[0098] Further, the charging box **900** may include a main control circuit board provided within the lower shell assembly **910** and electrode terminals **912** provided on the main control circuit board. The electrode terminals **912** may be provided in a plurality of sets, such as two sets, as required. The electrode terminals **912** may be exposed in the profiling recesses **911**, and when any one of the earphones **100** is put into the charging box **900**, the charging interface **16** in the earphone **100** may be in one-to-one contact with the electrode terminals **912** in the charging box **900** to meet the requirements of charging, detecting, and other functions. Correspondingly, the electrode terminals **912** may include a power positive terminal and a power negative terminal, and may further include a detection terminal; the connections between any two of the terminals may form a triangle, such as an equilateral triangle, or the terminals may be arranged in a spaced manner along a straight line, for example, spaced apart and collinear.

[0099] Alternatively, the charging box **900** may include a magnetic suction structure (not shown in the figures) provided within the lower shell assembly **910**. The magnetic suction structure may be provided in a plurality of sets, such as two sets, as required. After any of the earphones **100** is placed into the profiling recesses **911**, the magnetic suction structure and the magnetic suction element **17** within the earphone **100** may form a magnetic suction pairing to make the charging interface **16** and the electrode terminals **912** in one-to-one contact.

[0100] When the earphone **100** is in the wearing state, the first end **102** of the shell assembly **10** may need to extend into the user's concha cavity **502**. If the shape of the outer wall surface LS**3** of the first end **102** of the shell assembly **10** away from the user's ear cannot fit well with the concha cavity **502**, the earphone **100** may increase the foreign body sensation of the user during wearing, resulting in a poor experience.

[0101] In some embodiments, the first direction X has a positive direction pointing from the second end **103** to the first end **102**, and an outer wall surface LS**3** of the shell assembly **10** that is away from the user's ear when the earphone is in the wearing state may include a first arcuate transition region **105**. On the reference plane defined by the first direction X and the second direction Y, the first arcuate transition region **105** gradually approaches the user's ear along the positive direction of the first direction X, forming a first portion of the first end **102**. In such cases, an outer surface of the shell assembly **10** gradually bends towards the user's ear, which can better adapt to the shape of the concha cavity **502**, and reduce the foreign body sensation when worn by the user.

[0102] Referring to FIG. **7**, the radius of curvature of the first arcuate transition region **105** gradually reduces along the positive direction of the first direction X. In this case, the first curved transition region has a larger radius of curvature on a side away from the user's ear, which can be favorable for arranging elements inside the shell assembly **10**.

[0103] Considering that when the loudspeaker **13** and the battery **14** are arranged at an inclined angle, if the extension length of the first arcuate transition region **105** along the first direction X is too small, the size of a portion of the shell assembly **10** along the thickness direction may be relatively large, which reduces the utilization of the internal space of the shell assembly **10**. In some embodiments, an extension length L**105** of the first arcuate transition region **105** along the first direction X is not less than 16 mm. In such cases, the thickness of the shell assembly **10** at various locations can be within a reasonable range, and the utilization of the internal space of the shell assembly **10** can be improved.

[0104] As described above, along the first direction X, the first end **102** has the first reference point C**1** that is farthest away from the second end **103**, and the first reference point C**1** has been described above and will not be repeated herein. An angle α**1** between the tangent line of the first arcuate transition region **105** and the second direction Y at the first reference point C**1** may be in a range from 5 degrees and 8 degrees. For example, the angle α**1** between the first arcuate transition region **105** and the second direction Y at the first reference point C**1** is 5 degrees, 5.5 degrees, 6 degrees, 6.5 degrees, 7 degrees, 7.5 degrees, or 8 degrees. In some embodiments, see FIG. **7** and FIG. **10**, the first arcuate transition region **105** further has a first intermediate reference point C**4** whose distance L**14** from the first reference point C**1** along the first direction X is in a range from 1.8 mm and 3 mm. At the first intermediate reference point C**4**, an angle α**2** between the tangent line of the first arcuate transition region **105** and the second direction Y is in a range from 35 degrees and 53 degrees. In some embodiments, the first arcuate transition region **105** has a second intermediate reference point C**5** whose distance L**15** from the first reference point C**1** along the first

direction X is in a range from 6 mm and 9 mm. At the second intermediate reference point C**5**, an angle α**3** between the tangent line of the first arcuate transition region **105** and the second direction Y is in a range from 60 degrees and 80 degrees.
[0105] As mentioned above, by sequentially limiting the shapes of the first arcuate transition region **105** at the first reference point C**1**, the first intermediate reference point C**4**, and the second intermediate reference point C**5**, the first arcuate transition region **105** can effectively adapt to the shape of the user's concha cavity **502**.
[0106] In some specific embodiments, the distance L**14** between the first intermediate reference point C**4** and the first reference point C**1** along the first direction X is 2.35 mm. At the first intermediate reference point C**4**, the angle α**2** between the tangent line of the first arcuate transition region **105** and the second direction Y is 43.91 degrees. The distance L**14** between the second intermediate reference point C**5** and the first reference point C**1** along the first direction X is 7.43 mm, and the angle α**3** between the tangent line of the first arcuate transition region **105** and the second direction Y at the second intermediate reference point C**5** is 70.8 degrees.
[0107] In some embodiments, along the second direction Y, a side of the shell assembly **10** facing the user's ear has a third reference point C**3** that is adjacent to the first end **102** and closest to the user's ear, the outer wall surface LS**2** of the shell assembly **10** facing the user's ear has a second arcuate transition region **106** located between the first reference point C**1** and the third reference point C**3**, and the first direction X has a negative direction pointing from the first end **102** to the second end **103**. Along the negative direction of the first direction X, the second arcuate transition region **106** gradually approaches the user's ear, forming a second portion of the first end **102**. Optionally, the radius of curvature of the second arcuate transition region **106** gradually increases along the negative direction of the first direction X and gradually approaches the user's ear, thereby forming a second portion of the first end **102**.
[0108] In the above manner, the shape of the first end **102** can fit the shape of the interior of the concha cavity more closely, reducing the foreign body sensation when the first end **102** extends into the concha cavity **502**, and improving the wearing experience of the user.
[0109] In some embodiments, the distance L**13** between the third reference point C**3** and the first reference point C**1** along the first direction X is in a range from 3 mm and 6 mm. Optionally, at the third reference point C**3**, an angle α**4** between the tangent line of the second arcuate transition region **106** and the second direction Y is in a range from 75 degrees and 95 degrees.
[0110] In some specific embodiments, the distance L**13** between the third reference point C**3** and the first reference point C**1** along the first direction X is 4.29 mm, and the angle α**4** between the tangent line of the second arcuate transition region **106** and the second direction Y at the third reference point C**3** is 84.9 degrees.
[0111] In such cases, by limiting the projection length between the third reference point C**3** and the first reference point C**1** along the first direction X, the projection length between the third reference point C**3** and the first reference point C**1** along the second direction Y, and the slope of the second arcuate transition region **106** between the third reference point C**3** and the first reference point C**1**, the second arcuate transition region **106** can be better adapted to the interior of the concha cavity **502** and reduce foreign body sensation.
[0112] Further, the outer wall surface LS**2** of the shell assembly **10** facing the user's ear includes a third arcuate transition region **107** located on a side of the third reference point C**3** that is away from the first reference point C**1**, and the third arcuate transition region **107** is configured in an arch shape away from the user's ear. Along the first direction X, the distance L**10** between the apex G**1** of the third arcuate transition region **107** and the first reference point C**1** along the first direction X is in a range from 15 mm to 18 mm, and a distance H**10** between the apex G**1** of the third arcuate transition region **107** and the first reference point C**1** along the second direction Y is in a range from 1 mm to 3 mm.
[0113] In some specific embodiments, the distance L**10** between the apex G**1** of the third arcuate

transition region **107** and the first reference point C**1** along the first direction X is 17.7 mm, and the distance H**10** between the apex G**1** of the third arcuate transition region **107** and the first reference point C**1** along the second direction Y is 1.91 mm. In this case, the apex G**1** of the third arcuate transition region **107** is located substantially outside the tragus **508** of the user when the earphone **100** is in the wearing state, which can reduce the possibility of compressing the tragus **508**.

[0114] As shown in FIG. **11**, and as described above, the core module **1** may include the shell assembly **10**. The core module **1** may also include the loudspeaker **13** and the battery **14**, etc., as mentioned above.

[0115] Optionally, the shell assembly **10** may include an outer shell **11** and a decorative cover **12**, as shown in FIG. **11**. The outer shell **11** includes an outer surface away from the user's ear in a wearing state, and the decorative cover **12** is provided on the outer surface OS**1** of the outer shell **11** and is configured to form a touch positioning region **120** for the user to perform touch positioning. The core module **1** may further include a touch module **18**. The touch module **18** may be disposed in the outer shell **11** and includes a touch detection region **182** that at least partially overlaps the touch positioning region **120**. When the user touches the corresponding region on the decorative cover **12**, it can be detected by the touch detection region **182**.

[0116] The touch positioning region **120** may be used to position the touch detection region **182**. The user, upon sensing the touch positioning region **120**, can generally know that a region available for touch operation has been reached or is close. The touch detection region **182** may detect a touch operation of the user. The touch module **18** may detect the user's touch operation through the touch detection region **182**, thereby realizing touch sensing to enable the earphone **100** to realize the human-computer interaction function with the user. For example, functions such as volume adjustment, on/off control, song switching, play, and pause can be realized, and operations such as Bluetooth connecting or disconnecting operations, switching Bluetooth-connected devices, or the like can also be realized.

[0117] Further, the outer shell **11** and/or the decorative cover **12** may be provided with a positioning protrusion **121**, the positioning protrusion **121** being located within the touch positioning region **120** and protruding from the outer surface OS**1** of the decorative cover **12** that is away from the outer shell **11**. Based on the touch positioning region **120** providing a first touch navigation positioning, setting the positioning protrusion **121** to protrude from the outer surface OS**1** of the decorative cover **12** that is away from the outer shell **11** facilitates the user to touch the positioning protrusion **121** when touching the outer shell **11**, so that the positioning protrusion **121** can provide a second touch navigation positioning based on the decorative cover **12** to facilitate the user to more accurately locate the touch detection region **182** by touching the positioning protrusion **121**, improving the convenience of touch operation. Moreover, the decorative cover **12** can bring a decorative unique visual effect to the earphone **100**, and compounding (at least partially overlapping) with the touch detection region **182** based on the unique visual effect can eliminate the need to additionally set up a detection region for touch, making the earphone **100** smaller in size. Due to the visual effect of the decorative cover **12**, it is also easy for the user to be aware of the location of the touch detection region **182**, reducing the user's learning cost and improving the user's convenience of touch operation.

[0118] In some embodiments, as shown in FIG. **11**, the outer surface OS **1** of the outer shell **11** and an outer surface OS**2** of the decorative cover **12** may together form an outer surface OS of the core module **1**. The outer surface OS of the core module **1** is outwardly arch-shaped. In other words, the outer surface OS **1** of the outer shell **11** and the outer surface OS**2** of the decorative cover **12** are together outwardly arch-shaped, so that the lateral space utilization of the earphone **100** can be increased to dispose other components inside the shell assembly **10**, and the outwardly arch-shaped configuration can visually miniaturize the core module **1** on the outer surface OS.

[0119] In some embodiments, the first direction X and the second direction Y of the shell assembly **10** are likewise the first direction X and the second direction Y of the outer shell **11**, as shown in

FIG. **11**. The outer shell **11** has a length direction X, a width direction Z, and a thickness direction Y that are perpendicular to each other, and the outer shell **11** is configured to be stacked on the user's ear along the thickness direction Y in the wearing state. Specifically, the length direction X of the outer shell **11** may be as shown by the X arrow in FIG. **11**, the thickness direction Y of the outer shell **11** may be as shown by the Y arrow in FIG. **11**, and the width direction Z of the outer shell **11** may be as shown by the Z arrow in FIG. **11**, the direction X, the direction Y, and the direction Z being perpendicular to each other.

[0120] In some embodiments, as shown in FIG. **12**, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the length direction X may be greater than or equal to 0.5 and less than or equal to 0.8. The dimension of the decorative cover **12** along the length direction X may be as shown as a length D in FIG. **12**, and the dimension of the outer shell **11** along the length direction X may be as shown as a length E in FIG. **12**, where $0.5 \leq D/E \leq 0.8$. For example, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the length direction X may be 0.6, or 0.7, or the like. For example, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the length direction X is greater than or equal to 0.6 and less than or equal to 0.7.

[0121] Optionally, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the width direction Z may be greater than or equal to 0.6 and less than or equal to 0.95. The dimension of the decorative cover **12** along the width direction Z may be as shown as a length F in FIG. **12**, and the dimension of the outer shell **11** along the width direction Z may be as shown as a length G in FIG. **12**, where $0.6 \leq F:G \leq 0.95$. For example, the ratio of the dimension of the decorative cover **12** and the outer shell **11** along the width direction Z may be 0.65, or 0.75, etc. For example, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the width direction Z is greater than or equal to 0.7 and less than or equal to 0.87.

[0122] Setting the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** in this way can enable the decorative cover **12** to occupy a larger proportion of the outer surface OS**1** of the outer shell **11**, so that the decorative cover **12** can be easily touched by the user when touching the outer surface OS**1** of the outer shell **11**.

[0123] It should be appreciated that in other embodiments, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** may be other values, for example, the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the length direction X may be greater than or equal to 0.3 and less than or equal to 0.8, and the ratio of the dimension of the decorative cover **12** to the dimension of the outer shell **11** along the width direction Z may be greater than or equal to 0.4 and less than or equal to 0.9, which will not be specifically enumerated herein in this embodiment.

[0124] In some embodiments, as shown in FIG. **12**, the positioning protrusion **121** may be configured as a strip, with an extension direction H of the positioning protrusion **121** intersecting with the length direction X and the width direction Z, respectively. The extension direction H of the positioning protrusion **121** may be as shown by the H arrow in FIG. **12**. This setting enables the positioning protrusion, which intersects the length direction X and the width direction Z, respectively, to provide the user with an obvious tactile sensation when the user's hand is touching, which facilitates the user's touch positioning and provides a more obvious secondary touch navigation localization for the user.

[0125] Optionally, the extension direction H of the positioning protrusion **121** is not parallel to either the length direction X or the width direction Z of the outer shell **11**, and may intersect the length direction X of the outer shell **11** and form an acute angle with the length direction X. An angle a between the extension direction H of the positioning protrusion **121** and the length direction X of the outer shell **11** is shown in FIG. **12**.

[0126] Optionally, the angle between the extension direction H of the positioning protrusion **121** and the length direction X of the outer shell **11** may be in a range from 30 degrees to 70 degrees, e.g., the angle between the extension direction H of the positioning protrusion **121** and the length direction X of the outer shell **11** may be an angle of 40 degrees, 50 degrees, 60 degrees, or the like. Setting up in this way can make the positioning protrusion **121** easy to be touched and recognized by a user.

[0127] In some embodiments, as shown in FIGS. **12** and **14**, the outer shell **11** may include the connection end **103** (also referred to as the second end **103** in some descriptions of the present disclosure) connected to the ear hook **2** and the free end **102** (also referred to as the first end **102** in some descriptions of the present disclosure) that is away from the connection end **103** along the length direction X. The outer shell **11** may be bent along the thickness direction Y toward the direction in which the outer surface OS**1** faces, such that the free end **102** is closer to the user's ear hole than the connection end **103**, and the decorative cover **12** is closer to the connection end **103**. Since the free end **102** is closer to the ear and the connection end **103** connected to the ear hook **2** is farther away from the ear when the earphone **100** is worn by the ear, the decorative cover **12** is arranged closer to the connection end **103**, which is convenient for the user to touch the decorative cover **12**.

[0128] In some embodiments, as shown in FIG. **13** to FIG. **14**, the outer shell **11** may be bent along the thickness direction Y toward the direction in which the outer surface OS**1** of the outer shell **11** faces, so that the free end **102** is closer to the ear hole of the user's ear compared to the connection end **103**. With this configuration, the decorative cover **12** can be located, as much as possible, in a region parallel to the sagittal plane of the human body when the earphone **100** is worn, and as little as possible in the bent surface region extending toward the ear hole, which facilitates blind operation by the user and enables quick positioning of the positioning protrusion **121**.

[0129] In some embodiments, as shown in FIG. **13**, the free end **102** may include a first reference position K**1** that is farthest away from the connection end **103** along the length direction X, and the decorative cover **12** may include a second reference position K**2** that is closest to the first reference position K**1** along the length direction X. The free end **102** may extend at least partially into the concha cavity **502** in the wearing state.

[0130] The outer shell **11** includes an arcuate transition region **114** between the first reference position K**1** and the second reference position K**2**. From the second reference position K**2** to the first reference position K**1**, the arcuate transition region **114** is progressively closer to the user's ear along the thickness direction Y in the wearing state. As a result, when the earphone **100** is in the wearing state, a portion of the earphone **100** corresponding to the arcuate transition region **114** can gradually approach the user's ear or even fit to the user's ear along the thickness direction Y, thereby facilitating the extension of the free end **102** better into the concha cavity **502**, improving the wearing comfort and stability of the earphone **100**. Moreover, the arcuate transition region **114** starts from the second reference position K**2** on the decorative cover **12** and gradually inclines in an arc toward the first reference position K**1** such that the second reference position K**2** may serve as a turning point that helps the user more effectively locate the decorative cover **12**, thereby locating the touch positioning region **120** and enhancing the ease of touch interaction. In addition, the edge of the decorative cover **12** is close to the curve transition boundary of the outer shell **11**, making it easier for users to identify and locate the decorative cover **12** during blind operation.

[0131] In some embodiments, the roughness of the outer surface OS**2** of the decorative cover **12** may be greater than the roughness of the outer surface OS**1** of the outer shell **11**. When the user touches the decorative cover **12** and the outer surface OS**1** of the outer shell **11**, the difference in roughness can bring a difference in touch to the user, thereby enabling easy distinction between the decorative cover **12** and the outer shell **11**. As a result, the user can more easily locate the touch detection region **182** for touch input, improving the convenience of operation.

[0132] In some embodiments, the color of the outer surface OS**2** of the decorative cover **12** is different from the color of the outer surface OS**1** of the outer shell **11**. For example, the color of the outer surface OS**2** of the decorative cover **12** may be red, blue, etc., and the color of the outer surface OS**1** of the outer shell **11** may be white, black, etc., to allow the user to differentiate between the decorative cover **12** and the outer shell from the appearance. Setting the color of the outer surface OS**2** of the decorative cover **12** to be different from the color of the outer surface OS**1** of the outer shell **11** to produce a color difference on the outer surface OS of the core module **1** can visually make the core module **1** smaller in size, thereby enhancing the visual effect of the earphone **100** and visually minimizing the earphone **100**. Moreover, the color difference of the decorative cover **12** can enable visual positioning of the touch detection region **182**, enabling the user to conveniently know the location of the touch detection region **182**, reducing the cost of learning, and improving the efficiency of touch control.

[0133] In some embodiments, the material of the decorative cover **12** may be different from the material of the outer shell **11**. For example, the material of the decorative cover **12** may be a transparent acrylic sheet, and the material of the outer shell **11** may be metal. Or, the material of the decorative cover **12** may be metal, and the material of the outer shell **11** may be prepared from a material such as plastic, resin, or the like. The material of the decorative cover **12** being different from the material of the outer shell **11** can also make it easier for users to distinguish between the decorative cover **12** and the outer shell **11**, and the difference in materials can bring differentiated touch texture for the user, which is more convenient for the user's touch operation and improves the convenience of touch.

[0134] In some embodiments, at least a portion of the edge of the decorative cover **12** may be exposed to the outer surface OS**1** of the outer shell **11** to enclose the touch positioning region **120**, as illustrated in FIG. **14**. The touch positioning region **120** may be enclosed by at least a portion of the edge of the decorative cover **12** exposed to the outer surface OS**1** of the outer shell **11**. The user may touch the at least the portion of the edge of the decorative cover **12** when touching at the outer surface OS**1** of the outer shell **11**, and the presence of the edge provides tactile feedback, allowing the user to perceive that the corresponding region has been touched, thereby serving as an effective means of navigation and positioning.

[0135] Optionally, the edge of the decorative cover **12** may be configured as a ring and is completely exposed to the outer surface OS**1** of the outer shell **11** to enclose the touch positioning region **120**. The edge of the decorative cover **12** surrounds the positioning protrusion **121**. In other words, the edge of the decorative cover **12** is all exposed to the outer surface OS**1** of the outer shell **11**, so the user can touch the complete edge of the decorative cover **12**, which enhances the positioning effect and further improves the user's touch efficiency. Additionally, the positioning protrusion **121** is provided in the middle position of the decorative cover **12** and is not close to the edge of the decorative cover **12**, so as to make the positioning protrusion **121** more stable and not easy to detach from the touch positioning region **120** enclosed by the decorative cover **12**.

[0136] In some embodiments, as shown in FIG. **14**, the outer shell **11** may be provided with a positioning groove **113** on the outer surface, and the decorative cover **12** is accommodated in the positioning groove **113** through a groove opening of the positioning groove **113**.

[0137] Optionally, the shapes and sizes of the positioning groove **113** and the decorative cover **12** may be matched to facilitate the assembly of the decorative cover **12** in the positioning groove **113** of the outer shell **11**, which allows for a more even seam between the two, providing a more delicate and softer touch for the user.

[0138] Optionally, a seam between the edge of the decorative cover **12** and the edge of the groove opening of the positioning groove **113** is exposed to the outer surface, and the seam surrounds the positioning protrusion **121**. The seam is formed between the edge of the decorative cover **12** and the edge of the groove opening of the positioning groove **113**, which can provide a differentiated tactile sensation, and the user can perceive that the corresponding region has been touched, thereby improving the touch efficiency.

[0139] In some embodiments, the positioning protrusion **121** may protrude from the outer surface OS**1** of the outer shell **11**, the decorative cover **12** is provided with a positioning hole **122**, and the

positioning protrusion **121** passes through the positioning hole **122**. Through the positioning hole **122** and the positioning protrusion **121**, the decorative cover **12** and the outer surface OS**1** of the outer shell **11** can be accurately assembled, enabling a more accurate position of the touch detection region **182**. Furthermore, through the cooperation of the positioning groove **113** and the positioning protrusion **121**, the assembly accuracy between the decorative cover **12** and the outer shell **11** can be further enhanced, and the touch positioning region **120** can be positioned more accurately for the touch detection region **182**, thereby improving the accuracy and convenience of touch control.
[0140] Optionally, the positioning protrusion **121** may protrude from the outer surface OS**2** of the decorative cover **12** and be integrally molded with the decorative cover **12**, so that the positioning protrusion **121** can be made more stable and less likely to detach from the decorative cover **12**. Further, the relative position between the positioning protrusion **121** and the touch positioning region **120** remains fixed, allowing the user to achieve accurate primary and secondary touch-based navigation and positioning.
[0141] Optionally, at least two sound pickup holes **115** may be provided on the outer surface OS of the core module **1**. The two sound pickup holes **115** may be located on two opposite slopes of the outer surface OS of the core module **1**, respectively. Setting the two sound pickup holes **115** on two opposite slopes can enable the two sound pickup holes **115** to point in different directions, which expands the range of sound pickup of the earphone **100**, and pointing in different directions can minimize mutual interference between the two sound pickup holes **115**.
[0142] In some embodiments, the positioning protrusion **121** may be configured as a strip, with the two sound pickup holes **115** located on two sides of the positioning protrusion **121**, respectively. Setting the two sound pickup holes **115** on two sides of the positioning protrusion **121**, respectively, can minimize the interference between the two sound pickup holes **115**.
[0143] The positional relationship between the touch detection region **182** and the positioning protrusion **121** may be in various manners.
[0144] Manner 1: Optionally, the touch detection region **182** may enclose the positioning protrusion **121** to enable a user to accurately locate and confirm the position of the touch detection region **182** by touching the positioning protrusion **121**, which enables the user, when using the earphone **100**, to quickly and accurately locate the position of the touch detection region **182**, thereby optimizing the user experience.
[0145] Manner 2: The touch detection region **182** may be closer to the connection end **103** compared to the positioning protrusion **121**, so that the user can sense and trigger the touch detection region **182**. Or, the touch detection region **182** may surround the positioning protrusion **121** so that the user may precisely locate the touch detection region **182** and trigger the touch module **18** through the positioning protrusion **121**.
[0146] In some embodiments, as shown in FIG. **15** to FIG. **16**, the touch module **18** may include a touch metal pattern **181** and a circuit board **15**, the circuit board **15** may be disposed inside the outer shell **11**, and the touch metal pattern **181** may be disposed inside the outer shell **11** or sandwiched between the decorative cover **12** and the outer shell **11**. The touch metal pattern **181** is electrically connected to the circuit board **15**, and the touch metal pattern **181** is configured to form the touch detection region **182**.
[0147] When the user touches and/or presses the touch positioning region **120** or the positioning protrusion **121**, the touch metal pattern **181**, which forms the touch detection region **182**, may be triggered. The touch metal pattern **181** thus generates a corresponding signal and transmits the signal to the circuit board **15** to further initiate a response of other functions.
[0148] In some embodiments, the touch metal pattern **181** may also be configured as a radio frequency (RF) antenna of the earphone **100**. The earphone **100** may receive and transmit RF signals through the touch metal pattern **181**, so the functions of the earphone **100** can be extended and enriched to facilitate the earphone **100** to generate information interaction with other devices.
[0149] It should be understood that there may be at least one sound pickup hole **115**. The sound

pickup hole **115** at least passes through the outer shell **11**, and the accommodation space **101** is in flow communication with the external environment through the sound pickup hole **115**. Correspondingly, the core module **1** may further include a microphone **19**. The microphone **19** is disposed in the accommodation space **101** and corresponds to the sound pickup hole **115** to pick up sound from the external environment through the sound pickup hole **115**.

[0150] Optionally, as shown in FIGS. **17** to **20**, the core module **1** may be provided with at least two sound pickup holes **115** connecting the accommodation space **101** and the external environment. The core module **1** may further include at least two microphones **19**, the at least two microphones **19** being spaced apart in the accommodation space **101**, and corresponding to the at least two sound pickup holes **115** one by one. Taking the two microphones **19** as an example, one of the sound pickup holes **115** may pass through the outer shell **11** and be located at the periphery of the decorative cover **12**. The other one of the sound pickup holes **115** may pass through both the outer shell **11** and the decorative cover **12**, that is, the sound pickup hole **115** is located within the range enclosed by the edge of the decorative cover **12**. Correspondingly, the accommodation space **101** may be provided with two microphones **19**, corresponding to the two sound pickup holes **115** one by one.

[0151] By providing the at least two sound pickup holes **115**, the pickup range can be enlarged and the pickup quality can be improved. One of the sound pickup holes is disposed on the decorative cover **12**, and the other one of the sound pickup holes is disposed outside the decorative cover **12**, which can reduce the probability of blocking both of them when touching the decorative cover **12**, and thus reduce the impact on the sound picking effect.

[0152] Optionally, taking the two sound pickup holes **115** as an example, the distance between the two sound pickup holes **115** is greater than or equal to the dimension of the decorative cover **12** along the arrangement direction of the two microphones **19**. In such cases, since one of the sound pickup holes **115** is disposed outside the decorative cover **12**, the other one of the sound pickup holes **115** can be located at the edge of the decorative cover **12**, so that the distance between the two sound pickup holes **115** is relatively large. In this way, the sound pickup hole **115** located outside the decorative cover **12** is not easily blocked when the touching positioning region **120** of the decorative cover **12** is touched, and the sound pickup hole **115** located close to the edge of the decorative cover **12** is not easily blocked as well, which can further minimize the probability of blocking the two sound pickup holes **115** at the same time when touching the decorative cover **12**, and thus reduce the impact on the sound pickup effect.

[0153] Optionally, as shown in FIG. **17**, the touch positioning region **120** is located between the two sound pickup holes **115**. Set up in this way, it is not easy to block the two sound pickup holes **115** when touching the touch positioning region **120**, which reduces the impact on the sound pickup effect.

[0154] Optionally, the arrangement direction of the two sound pickup holes **115** intersects the length direction X and the width direction Z of the outer shell **11**, respectively. The sound pickup hole **115** that passes through the decorative cover **12** (i.e., the sound pickup hole **115** located within the range enclosed by the decorative cover **12**) is closer to the connection end **103** and closer to the edge of the decorative cover **12** that is away from the ear hook **2** along the width direction Z, compared to the sound pickup hole **115** located at the periphery of the decorative cover **12**. The sound pickup hole **115** is disposed toward the lower left side, which, on the one hand, allows the sound pickup hole **115** that passes through the decorative cover **12** to be closer to the user's mouth and the loudspeaker **13**, facilitating clearer capture of the user's voice and the sound from the loudspeaker **13**, thereby enabling more effective noise reduction processing and improving the pickup quality of ambient sounds. On the other hand, the sound pickup hole **115** is closer to the edge of the decorative cover **12** that is away from the ear hook **2** along the width direction Z, which reduces the probability of the sound pickup hole **115** being blocked when the decorative cover **12** is touched.

[0155] Optionally, as shown in FIGS. **18** to **20**, the core module **1** may further include a sealing component **190**, at least a portion of the sealing component **190** being disposed between the outer shell **11** and the microphone **19** for isolating a sound pickup path from the accommodation space **101**, such that the sound pickup path is sealed relative to the accommodation space **101** located outside the sound pickup path, which helps prevent dust, moisture, and other contaminants from affecting other electronic components in the accommodation space **101** after entering through the sound pickup hole **115** into the sound pickup path.
[0156] Optionally, the core module **1** may further include a fixing plate for fixing the microphone **19**. Optionally, the fixing plate may be the circuit board **15**. Or, in other embodiments, the fixing plate may be a board-like structural member other than the circuit board **15**, or a portion of the shell assembly **10**, which may be used to fix the microphone **19**, and may further fix other components.
[0157] As shown in FIGS. **18** to **20**, the fixing plate is accommodated in the accommodation space **101** and includes a through hole **151**, and the microphone **19** is fixed to one side of the fixing plate and covers the through hole **151**. At least a portion of the sealing component **190** is disposed in the accommodation space **101** and abuts between the outer shell **11** and a side of the fixing plate away from the microphone **19**, and enables the through hole **151** and the sound pickup hole **115** to be in flow communication with each other. Such a setting can improve the stability and sealing performance of the sealing component **190**.
[0158] There are a variety of mating configurations for the sealing component **190** with the outer shell **11**, etc., which can be tailored to different mounting configurations or different locations of the microphone **19**.
[0159] In some embodiments, for the sound pickup hole **115** that passes through the outer shell **11** and is located at the periphery of the decorative cover **12**, an exemplary structure of the sealing component **190** and relevant descriptions are as follows.
[0160] As shown in FIGS. **18** and **19**, the microphone **19** may pass through the outer shell **11** and be located at the periphery of the decorative cover **12**. The sealing component **190** may include a sealing gasket **191***a* and a mesh component **192***a*, the sealing gasket **191***a* and the mesh component **192***a* being stacked and disposed in the accommodation space **101** and located between the fixing plate and the outer shell **11**. Since the microphone **19** is located at the periphery of the decorative cover **12** and passes through the outer shell **11**, the outer shell **11** is more monolithic and integrally formed, therefore, the sealing gasket **191***a* and the mesh component **192***a* are stacked and disposed between the fixing plate and the outer shell **11**, on the one hand, the sealing effect can be better realized due to the outer shell **11** with better integrity, on the other hand, the stacking of sealing gasket **191***a* and the mesh component **192***a* can realize a thicker sealing component **190**, which can be tightly disposed between the outer shell **11** and the fixing plate. Besides, the mesh component **192***a* can prevent dust and impurities from entering the accommodation space **101**, further improving the sealing effect.
[0161] Optionally, the sealing gasket **191***a* is stacked on the fixing plate and is closer to the through hole **151** as compared to the mesh component **192***a*. The mesh component **192***a* is disposed between the sealing gasket **191***a* and the outer shell **11** and is closer to the sound pickup hole **115** compared to the sealing gasket **191***a*. Thus, placing the mesh component **192***a* in front of the sealing gasket **191***a* (i.e., closer to the sound pickup hole **115**) can prevent dust, impurities, etc. from entering into and building up in the sealing gasket **191***a*, which allows dust, impurities, and moisture to be blocked at an earlier stage, thereby minimizing their entry into the accommodation space **101**.
[0162] In other embodiments, for the sound pickup hole **115** that passes through the outer shell **11** and the decorative cover **12**, an exemplary structure of the sealing component **190** and related descriptions are as follows.
[0163] As shown in FIG. **20**, the microphone **19** may pass through the outer shell **11** and the decorative cover **12**, and the sealing component **190** may include a sealing gasket **191***b* and at least

one mesh component **192***b*. The sealing gasket **191***b* is disposed in the accommodation space **101** and i between the outer shell **11** and the fixing plate, and directly abuts the outer shell **11** and the fixing plate. The at least one mesh component **192***b* is disposed between the outer shell **11** and the decorative cover **12**. Due to the relatively large tolerance that may accumulate from the fit between the decorative cover **12** and the outer shell **11**, as well as from the fit between the sealing component **190** and the fixing plate, the sealing gasket **191***b* with stronger deformation ability is connected to the fixing plate, which helps reduce the fit tolerance between the decorative cover **12** and the outer shell **11**, and between the sealing component **190** and the fixing plate, thereby ensuring the sealing performance of the circuit components.
[0164] Optionally, the at least one mesh component **192***b* includes two mesh components, with one of the two mesh components **192***b* disposed between the outer shell **11** and the decorative cover **12**, and the other one of the two mesh components **192***b* stacked with the sealing gasket **191***b* in the accommodation space **101** and disposed between the outer shell **11** and the fixing plate.
[0165] By disposing the mesh component **192***b* between the outer shell **11** and the decorative cover **12**, on the one hand, dust, impurities, and moisture can be reduced from entering the accommodation space **101** at an earlier stage, on the other hand, the mesh component **192***b* also serves to seal the sound pickup hole **115** between the outer shell **11** and the decorative cover **12**, thereby reducing the ingress of moisture through the sound pickup hole into the space between the outer shell **11** and the decorative cover **12**, which helps maintain the stability of their connection. By disposing the sealing gasket **191***b* inside the accommodation space **101** and sealing between the outer shell **11** and the fixing plate, the sealing performance is further enhanced, thereby reducing the leakage of moisture and other contaminants entering through the sound pickup hole **115** into the accommodation space **101**, which could otherwise adversely affect the internal electronic components.
[0166] For the mesh components **192***a* and **192***b*, the exemplary descriptions of their structure are as follows.
[0167] As shown in FIG. **19** to FIG. **20**, the mesh components **192***a* and **192***b* include an annular double-sided adhesive **1921** and a mesh **1922**. The annular double-sided adhesive **1921** is stacked with the mesh **1922** and bonded to the mesh **1922**. Specifically, one side of the annular double-sided adhesive **1921** may be bonded to the mesh **1922**, and the other side may be connected to the outer shell **11** or the decorative cover **12** close to it. For example, for the sound pickup hole that passes through the outer shell **11** and the decorative cover **115**, one side of the annular double-sided adhesive **1921** may be bonded to the mesh **1922**, and the other side may be bonded to the outer shell **11**. As another example, for the sound pickup hole **115** that passes through the outer shell **11** and is located at the periphery of the decorative cover **12**, one side of the annular double-sided adhesive **1921** may be bonded to the mesh **1922**, and the other side may be bonded to the outer shell **11**. The mesh **1922** may face the sound pickup hole **115** and the through hole **151** to efficiently prevent impurities, dust, or the like from entering the accommodation space **101**. Specifically, the annular double-sided adhesive **1921** surrounds the periphery of the sound pickup hole **115**, and a portion of the mesh **1922** disposed in the middle of the annular double-sided adhesive **1921** faces the sound pickup hole **115**.
[0168] Optionally, the least one mesh component **192***a* includes two layers of annular double-sided adhesive **1921** and one layer of mesh **1922**, and the mesh **1922** is stacked and bonded between the two layers of annular double-sided adhesive **1921**. By disposing the two layers of annular double-sided adhesive **1921**, it's the bonding strength of the mesh **1922** can be enhanced, the mesh **1922** can be better fixed, and the thickness of the annular double-sided adhesive **1921** can be increased, so that the annular double-sided adhesive **1921** can be more tightly sealed between the outer shell **11** and the decorative cover **12**, or between the outer shell **11** and the sealing gaskets **191***a* and **191***b*, thereby improving the sealing effect.
[0169] Optionally, the at least one mesh component **192***a* includes two layers of annular double-

sided adhesive **1921** and two layers of mesh **1922**, the two layers of annular double-sided adhesive **1921** are spaced apart, one layer of the mesh **1922** is stacked and bonded between the two layers of annular double-sided adhesive **1921**, and one layer of the annular double-sided adhesive **1921** is stacked between the two layers of mesh **1922**. By disposing the two layers of mesh **1922**, the filtration and blocking effect can be further enhanced, and by disposing the two layers of annular double-sided adhesive **1921**, the two layers of mesh **1922** can be effectively fixed with an increased thickness, which can be more tightly sealed between the outer shell **11** and the decorative cover **12**, or between the outer shell **11** and the sealing gaskets **191***a* and **191***b*, thereby improving the sealing effect.

[0170] In conjunction with FIG. **21** to FIG. **23**, the foregoing content has illustrated that the earphone **100** may include the shell assembly **10** and the loudspeaker **13**. Optionally, the earphone **100** may further include a sealant (not shown in the figures). The shell assembly **10** is configured to form a first accommodation cavity **1011**. The loudspeaker **13** includes a basket **131** and a metal pad **132**. In conjunction with FIG. **24** and FIG. **25**, the basket **131** is disposed in the first accommodation cavity **1011**, and the metal pad **132** is disposed on the basket **131** and faces the battery **14** to ensure that the loudspeaker **13** is electrically connected to other elements. The sealant is configured to seal the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**.

[0171] By disposing the sealant to seal the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, a seal is formed between the loudspeaker **13** and the shell assembly, thereby enhancing the sealing performance of the shell assembly **10** and reducing the risk of water ingress damaging the electronic elements.

[0172] When dispensing the sealant, it is often necessary to dispense the sealant around the basket **131**, so that the sealant can be continuously filled between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, thereby achieving a stable and reliable sealing effect between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**. In the prior art, the metal pad **132** is usually disposed to extend from the basket **131** to the outside of the inner peripheral wall of the first accommodation cavity **1011** to connect to the power source outside the first accommodation cavity **1011**, and when dispensing the sealant, the metal pad **132** tends to significantly block the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, making it difficult to dispense the sealant near the metal pad **132**, which may lead to discontinuity of the sealant at the metal pad **132**, thereby reducing the sealing reliability.

[0173] By disposing the metal pad **132** on the side of the inner peripheral wall of the first accommodation cavity **1011** facing the battery **14**, and by setting the metal pad **132** to be spaced apart from the inner peripheral wall of the first accommodation cavity **1011**, the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011** at the metal pad **132** during dispensing can be exposed to the outside world, thereby facilitating dispensing the sealant at the metal pad **132** and enabling the sealant to be continuously filled between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, thereby realizing stable and reliable sealing effect.

[0174] Further, a wire **133** may extend from the metal pad **132** to the inner peripheral wall of the first accommodation cavity **1011**, the diameter of the wire **133** is much smaller than the dimension of the metal pad **132**, and the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011** is less blocked by the wire **133**, which is conducive to dispensing the sealant and achieving a stable and reliable sealing effect.

[0175] In conjunction with FIG. **23** to FIG. **25**, in some embodiments, the metal pad **132** is disposed not to extend beyond the outer peripheral wall of the basket **131**.

[0176] Set up in this way, during the dispensing process, the metal pad **132** may not block the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, which facilitates dispensing the sealant and realizing a stable and reliable sealing effect.

[0177] In conjunction with FIG. **23**, FIG. **24**, and FIG. **26**, in some embodiments, the metal pad **132** is disposed to extend beyond an outer peripheral wall of the basket **131**, and an excess distance S**1** of the metal pad **132** relative to the outer peripheral wall of the basket **131** is not greater than one-half of a dimension S**2** of the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**.

[0178] Set up in this way, during the dispensing process, the metal pad **132** may not completely block the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, and the dispensing operation can still be performed at the metal pad **132**. By setting the excess distance S**1** of the metal pad **132** relative to the outer peripheral wall of the basket **131** being not greater than one-half of the dimension S**2** of the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, during the dispensing operation, the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011** at the metal pad **132** can be partially exposed to the outside world. Since the sealant has a certain mobility during dispensing, the mobility of the sealant can be utilized to make the sealant fill the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**, thereby realizing a stable and reliable sealing effect. For example, the excess distance S**1** of the metal pad **132** relative to the outer peripheral wall of the basket **131** may be set to be 0.2 to 0.3 times, or 0.4 to 0.5 times, of the dimension S**2** of the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**.

[0179] In some embodiments, in conjunction with FIG. **23**, FIG. **27**, and FIG. **28**, the first accommodation cavity **1011** has an open end **1014**, the basket **131** is disposed within the first accommodation cavity **1011** through the open end **1014**, the metal pad **132** is disposed on one side of the basket **131** facing the open end **1014**, and the earphone **100** further includes the wire **133**, the wire **133** being soldered to a side of the metal pad **132** facing the open end **1014**.

[0180] During the assembly process of the basket **131**, the peripheral side of the basket **131** and a side of the basket **131** away from the open end **1014** may be in contact with the inner wall of the first accommodation cavity **1011**, and by setting the metal pad **132** on the side of the basket **131** facing the open end **1014** and soldering the wire **133** to the side of the metal pad **132** facing the open end **1014**, the interference of the metal pad **132** and the wire **133** on the assembly process can be reduced and the assembly efficiency can be improved.

[0181] In conjunction with FIG. **23** to FIG. **25**, in some embodiments, the sealant is applied from the side of the open end **1014** to the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011**.

[0182] The gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011** may be exposed to the outside world through the open end **1014** during the dispensing process, thereby facilitating dispensing the sealant into the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011** through the open end **1014** from the outside world, such that the sealant enters the gap between the outer peripheral wall of the basket **131** and the inner peripheral wall of the first accommodation cavity **1011** from the side of the open end **1014**. Set up in this way, it is convenient to perform the dispensing operation and improve the assembly efficiency.

[0183] In combination with FIG. **27** to FIG. **29**, in some embodiments, the earphone **100** further includes a bracket **134**. The basket **131** is provided with an annular table surface **1311** facing the open end **1014** and an assembly table surface **1312**, the annular table surface **1311** being disposed along a circumferential direction of the basket **131** and provided with a first notch **1313**, and the assembly table surface **1312** being disposed along a circumferential direction of the basket **131** and corresponding to the first notch **1313**. The metal pad **132** is disposed on the assembly table surface **1312**, and the bracket **134** is supported on the annular table surface **1311**.

[0184] The assembly table surface **1312** may be sunk in a direction away from the open end **1014** relative to the annular table surface **1311** at the first notch **1313**. The metal pad **132** has a raised height. By providing the annular table surface **1311** with the first notch **1313**, the bracket **134** can avoid the metal pad **132** as well as the wire **133** when the bracket **134** is supported on the annular table surface **1311**, thereby facilitating an assembly of the bracket **134** with the basket **131** and improving assembly efficiency.

[0185] In some embodiments, in conjunction with FIG. **28** and FIG. **29**, the bracket **134** is provided with a second notch **1341** at an end facing the basket **131**, the second notch **1341** corresponding to the assembly table surface **1312** along the circumferential direction of the basket **131**.

[0186] By providing the second notch **1341**, the bracket **134** can avoid the metal pad **132** and the wire **133** more easily when supported on the annular table surface **1311**, facilitating the assembly of the bracket **134** with the basket **131** and improving the assembly efficiency.

[0187] In some embodiments, in conjunction with FIG. **22** to FIG. **24**, the shell assembly **10** further forms a second accommodation cavity **1012** and includes the partition wall **104** for separating the first accommodation cavity **1011** and the second accommodation cavity **1012**. The earphone **100** further includes the battery **14** disposed in the second accommodation cavity **1012**, and the metal pad **132** is disposed on a side of the basket **131** facing the partition wall **104**.

[0188] A portion of the inner peripheral wall of the first accommodation cavity **1011** may be formed by the partition wall **104**, and a portion of the sealant may seal the gap between the outer peripheral wall of the basket **131** and the partition wall **104**. By providing the partition wall **104**, the risk of water flowing from a side of the loudspeaker **13** close to a sound outlet hole **1013** to the second accommodation cavity **1012** may be reduced.

[0189] The battery **14** may be used as a power source to power the loudspeaker **13**. The first accommodation cavity **1011** and the second accommodation cavity **1012** are provided adjacent to each other, which is conducive to simplifying the connection line between the battery **14** and the loudspeaker **13**. By disposing the metal pad **132** on the side of the basket **131** facing the partition wall **104**, the spacing distance between the battery **14** and the metal pad **132** can be reduced, which is conducive to further simplifying the connection line between the battery **14** and the loudspeaker **13**. Further, the metal pad **132** may be connected to the battery **14** via the wire **133** to enable the battery **14** to power the loudspeaker **13**.

[0190] The foregoing descriptions are merely part of the embodiments of the present disclosure and are not intended to limit the scope of protection of the present disclosure. Any equivalent devices or equivalent process modifications made based on the contents of the specification and drawings of the present disclosure, or any direct or indirect applications in other related technical fields, shall be deemed to fall within the scope of protection of the present disclosure.

Claims

1. An earphone, comprising: an ear hook; and a core module connected to the ear hook, the core module including an outer shell, a decorative cover, and a touch module, the outer shell including an outer surface away from an ear of a user when the earphone is in a wearing state, the decorative cover being disposed on the outer surface of the outer shell and configured to form a touch positioning region for the user to perform touch positioning; a positioning protrusion being disposed on the outer shell and the decorative cover, the positioning protrusion being located in the touch positioning region and protruding from an outer surface of the decorative cover away from the outer shell; the touch module being disposed on the outer shell and including a touch detection region that at least partially overlaps with the touch positioning region.

2. The earphone of claim 1, wherein the edge of the decorative cover is configured as a ring and is completely exposed on the outer surface of the outer shell to enclose the touch positioning region; and the edge of the decorative cover surrounds the positioning protrusion.

3. The earphone of claim 1, wherein a positioning groove is disposed on the outer surface of the outer shell, and the decorative cover is accommodated in the positioning groove through a groove opening of the positioning groove the positioning groove and the decorative cover match in shape and size; and a seam between an edge of the decorative cover and an edge of the groove opening of the positioning groove is exposed on the outer surface, and the seam surrounds the positioning protrusion.

4. The earphone of claim 1, wherein the positioning protrusion protrudes from the outer surface of the outer shell, the decorative cover is provided with a positioning hole, and the positioning protrusion passes through the positioning hole; or, the positioning protrusion protrudes from the outer surface of the decorative cover and is integrally formed with the decorative cover.

5. The earphone of claim 1, wherein the outer shell has a length direction, a width direction, and a thickness direction that are perpendicular to each other, and the outer shell is stacked on the ear of the user along the thickness direction when the earphone is in the wearing state; a ratio of a dimension of the decorative cover to a dimension of the outer shell along the length direction is greater than or equal to 0.5 and less than or equal to 0.8; and a ratio of a dimension of the decorative cover to a dimension of the outer shell along the width direction is greater than or equal to 0.6 and less than or equal to 0.95.

6. The earphone of claim 5, wherein the outer shell includes a connection end connected to the ear hook and a free end away from the connection end along the length direction, and the decorative cover is closer to the connection end.

7. The earphone of claim 6, wherein the outer shell is bent along the thickness direction toward the outer surface of the outer shell so that the free end is closer to an ear hole of the ear of the user than the connection end.

8. The earphone of claim 1, wherein the outer shell has a length direction, a width direction, and a thickness direction that are perpendicular to each other, and the outer shell is stacked on the ear of the user along the thickness direction when the earphone is in the wearing state; and the positioning protrusion is configured as a strip, and an extension direction of the positioning protrusion intersects with the length direction and the width direction, respectively.

9. The earphone of claim 1, wherein the touch module includes a touch metal pattern and a circuit board, the circuit board is disposed inside the outer shell, the touch metal pattern is disposed inside the outer shell or sandwiched between the decorative cover and the outer shell, the touch metal pattern is electrically connected to the circuit board, and the touch metal pattern is configured to form the touch detection region.

10. The earphone of claim 1, wherein the outer shell has a length direction, a width direction, and a thickness direction that are perpendicular to each other, and the outer shell is stacked on the ear of the user along the thickness direction when the earphone is in the wearing state; the outer shell includes a connection end connected to the ear hook and a free end away from the connection end along the length direction, and the outer shell is bent along the thickness direction toward the outer surface of the outer shell, so that the free end is closer to an ear hole of the ear of the user than the connection end, and the decorative cover is closer to the connection end.

11. The earphone of claim 10, wherein the outer surface of the outer shell and the outer surface of the decorative cover together form an outer surface of the core module; the outer surface of the core module is configured as an outwardly arched shape; at least two sound pickup holes are disposed on the outer surface of the core module; the two sound pickup holes are located on two opposite slopes of the outer surface of the core module, respectively; and the touch detection region is closer to the connection end than the positioning protrusion, or the touch detection region surrounds the positioning protrusion.

12. The earphone of claim 11, wherein the positioning protrusion is configured as a strip, and the two sound pickup holes are located on two sides of the positioning protrusion, respectively.
13. The earphone of claim 10, wherein the free end has a first reference position farthest from the connection end along the length direction, and the decorative cover has a second reference position closest to the first reference position along the length direction; the outer shell includes an arcuate transition region between the first reference position and the second reference position; and from the second reference position to the first reference position, the arcuate transition region gradually approaches the ear of the user along the thickness direction when the earphone is in the wearing state.
14. The earphone of claim 1, wherein the outer shell forms an accommodation space, and the core module is provided with two sound pickup holes, the accommodation space is in flow communication with the external environment through the two sound pickup holes, one of the two sound pickup holes passes through the outer shell and is located at a periphery of the decorative cover, and the other one of the two sound pickup holes passes through the outer shell and the decorative cover; and the core module includes two microphones, and the two microphones are spaced apart in the accommodation space and correspond one-to-one with the two sound pickup holes.
15. The earphone of claim 14, wherein a distance between the two sound pickup holes is greater than or equal to a dimension of the decorative cover along an arrangement direction of the two microphones; the outer shell has a length direction and a width direction that are perpendicular to each other, the outer shell has a connection end and a free end along the length direction; the connection end is connected to the ear hook; the arrangement direction of the two microphones intersects with the length direction and the width direction, respectively; and the sound pickup hole that passes through the decorative cover is closer to the connection end and an edge of the decorative cover away from the ear hook along the width direction than the sound pickup hole located at the periphery of the decorative cover; and, the touch detection region is located between the two sound pickup holes.
16. The earphone of claim 1, wherein the outer shell forms an accommodation space, and the core module is provided with a sound pickup hole passing through the outer shell, the accommodation space is in flow communication with the external environment though the sound pickup hole; the core module includes a microphone and a sealing component, and the microphone is disposed in the accommodation space and corresponds to the sound pickup hole to pick up sound of the external environment through the sound pickup hole; and at least a portion of the sealing component is disposed between the outer shell and the microphone and is configured to isolate a sound pickup path from the accommodation space.
17. The earphone of claim 16, wherein the core module further includes a fixing plate, the fixing plate is accommodated in the accommodation space and includes a through hole, and the microphone is fixed to one side of the fixing plate and covers the through hole; and at least a portion of the sealing component is disposed in the accommodation space and abuts between the outer shell and a side of the fixing plate away from the microphone, and enables the through hole and the sound pickup hole to be in flow communication with each other.
18. The earphone of claim 17, wherein the microphone passes through the outer shell and is located at a periphery of the decorative cover; the sealing component includes a sealing gasket and a mesh component, and the sealing gasket and the mesh component are stacked and disposed in the accommodation space between the fixing plate and the outer shell.
19. The earphone of claim 18, wherein the sealing gasket is stacked on the fixing plate and is closer to the through hole than the mesh component; and the mesh component is located between the sealing gasket and the outer shell and is closer to the sound pickup hole than the sealing gasket.
20. The earphone of claim 17, wherein the microphone passes through the outer shell and the decorative cover; and the sealing component includes a sealing gasket and at least one mesh

component, the sealing gasket is disposed in the accommodation space between the outer shell and the fixing plate and directly abuts the outer shell and the fixing plate, and the least one mesh component is disposed between the outer shell and the decorative cover.